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# Applied Data Science Project

L2 - Project pillars

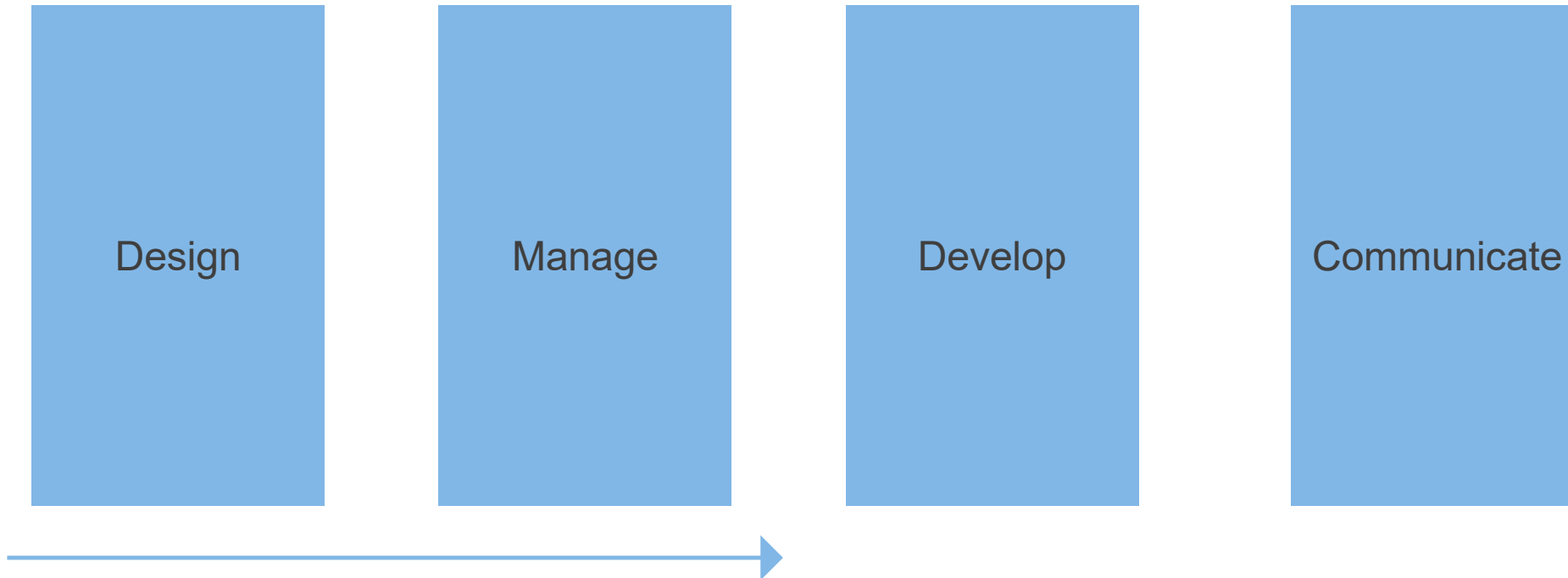


**Politecnico  
di Torino**



**e11is**  
European Laboratory for Learning and Intelligent Systems

# Pillars



Executed sequentially, meaning the development comes after Design and Manage

Develop and Communicate may be executed in parallel, usually after the preparedness of the project, namely Design and Manage

# Design

## Pillar 1

A project design starts with the definition of the **objectives**, **outcomes** and **output**

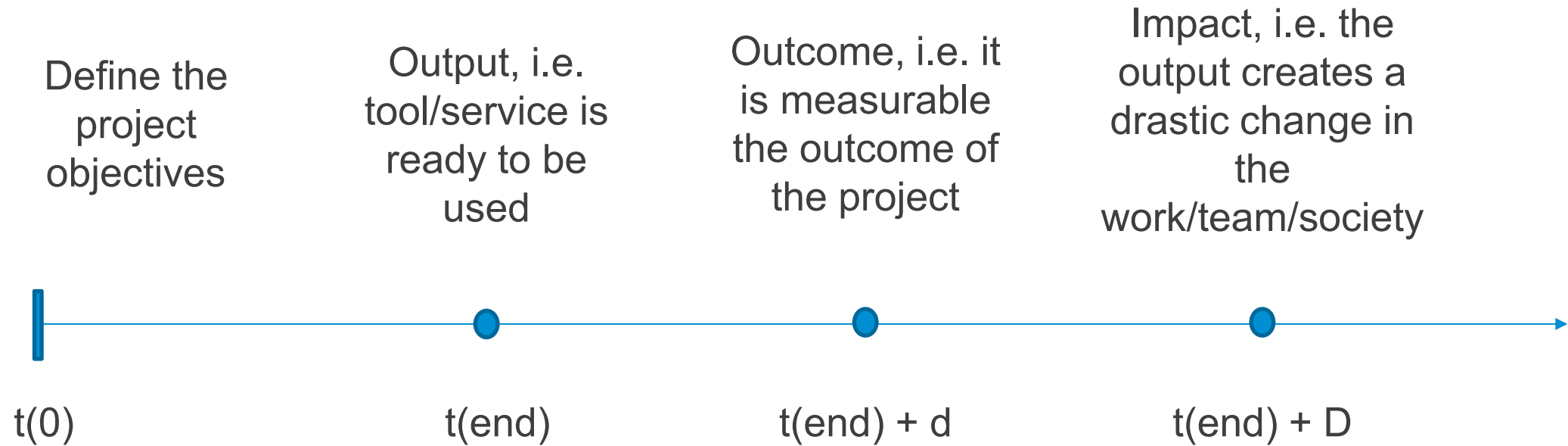
An **output** is a software program, or a training material, or ... a workshop or a stroller

An **outcome** is the difference that the output makes if compared with the existing situation, for instance the software program helps the team to work faster, the stroller to keep a baby calm and relaxed

An **objective** is the description of a goal, for instance lowering poverty or increasing societal inclusion

# Project timeline

## Pillar 1



$$d \ll D$$

# Centered around the user

## Pillar 1

The design involves the identification of user requirements

Requirements are translated in journeys that are use cases that underline the user pains and pave the way to outline the output

A persona is a fictional character utilized to represent a user type that will use the product or the solution being generated

A fictional character has both **needs** and **frustrations** and requires **tools** to address a situation

# Customer journey map

## Pillar 1

It is defined as a comprehensive approach for improving the experience customers have along the service journey

### Online Shopping Customer Journey Map

| Stages of Journey     | Motivation                                                                        | Searches for Websites                                                                                               |                                                                                    |                                                                                                                                                    | Browses the Site                                                                    | Evaluates Products                                                                   |                                                                                                                                 | Pay                                                                                  |                                                                                                                                            |
|-----------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Activities            | Wants to buy a Christmas gift for a friend                                        | Searches products keywords on search engines                                                                        | Clicks the 1st ads. result.                                                        | Checks online again and go to the 1st. original result                                                                                             | Checks out ongoing deals and hot sale products                                      | Opens a product page to check product details.                                       | Opens more product pages to compare info.                                                                                       | Processes to pay a preferred product.                                                | Contacts online service for help.                                                                                                          |
| Feelings              |  |                                    |   |                                                                 |  |   |                                              |   |                                                         |
| Very Happy            |  |                                                                                                                     |                                                                                    |                                                                                                                                                    |  |                                                                                      |                                                                                                                                 |                                                                                      |                                                                                                                                            |
| Overall Satisfied     |                                                                                   |                                    |                                                                                    |                                                                 |                                                                                     |                                                                                      |                                              |                                                                                      |                                                         |
| Unhappy               |                                                                                   |                                                                                                                     |  |                                                                                                                                                    |                                                                                     |  |                                                                                                                                 |  |                                                                                                                                            |
| Experiences           | Excited to give a surprise to the friend.                                         | <ul style="list-style-type: none"> <li>Happy to see many options</li> <li>Confusing about how to choose.</li> </ul> | Annoyed at the useless info.                                                       | <ul style="list-style-type: none"> <li>Happy with the informative shopping website.</li> <li>Not sure about how to find the best price.</li> </ul> | Surprise to see lots of discounts                                                   | Frustrated about reviews from other customers.                                       | <ul style="list-style-type: none"> <li>Pleased with usages &amp; designs.</li> <li>Sad to see out of stock products.</li> </ul> | Frustrated to find only the Paypal payment option.                                   | <ul style="list-style-type: none"> <li>Satisfied with bank card payment.</li> <li>Feels unhappy to wait for a bit of long time.</li> </ul> |
| Customer Expectations | Easily to obtain discount news.                                                   | More user-friendly search engine.                                                                                   | Allow to hide unnecessary ads.                                                     | Clear and innovative website design.                                                                                                               | More discounts during holidays.                                                     | Higher quality products.                                                             | Faster supply chain system.                                                                                                     | More payment choices.                                                                | Faster website speed.                                                                                                                      |

# Personas

## Pillar 1

| Persona One                                                                       | Persona Two                                                                         | Persona Three                                                                       |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |
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# Functional requirements

## Pillar 1

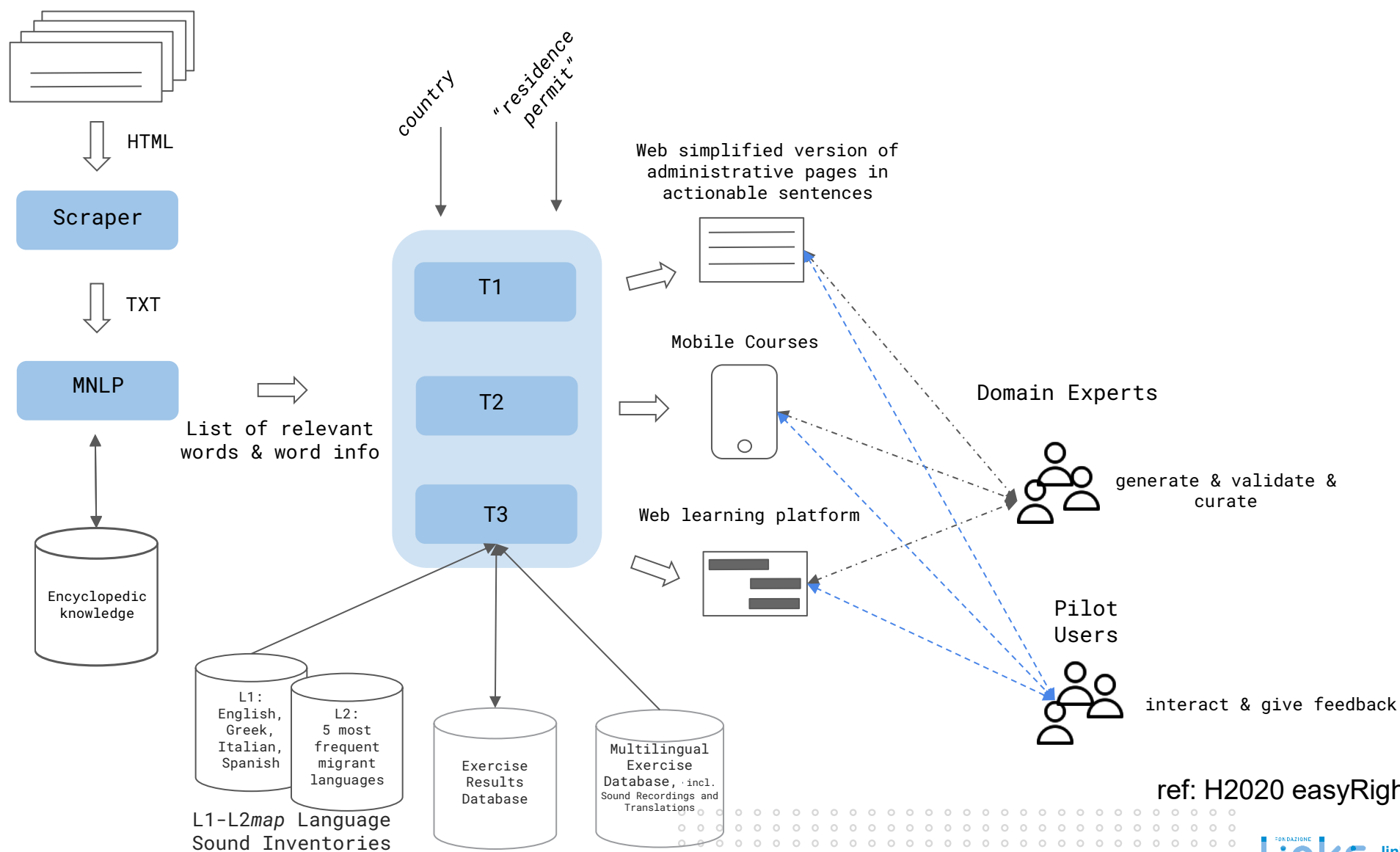
Needs of the fictional characters are translated into requirements of the product or solution

| No. | Requirement                                                                                     | Notes | Category  |
|-----|-------------------------------------------------------------------------------------------------|-------|-----------|
| 1   | Students should be able to use a live chatbox functionality to connect with available tutors.   | N/A   | Technical |
| 2   | Complete a roll-out of the portal to 30 after-school clubs in the state.                        | N/A   | Business  |
| 3   | The portal should include a section for explainer videos across four core educational subjects. | N/A   | Technical |

Example of a table that lists the requirements

# Functional diagram

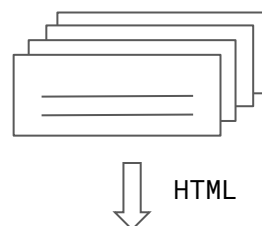
## Pillar 1



ref: H2020 easyRights project

# Functional diagram

## Pillar 1

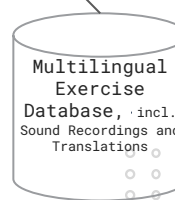


country  
"residence permit"

Web simplified version of administrative pages in actionable sentences

Whole project

Encyclopedic knowledge

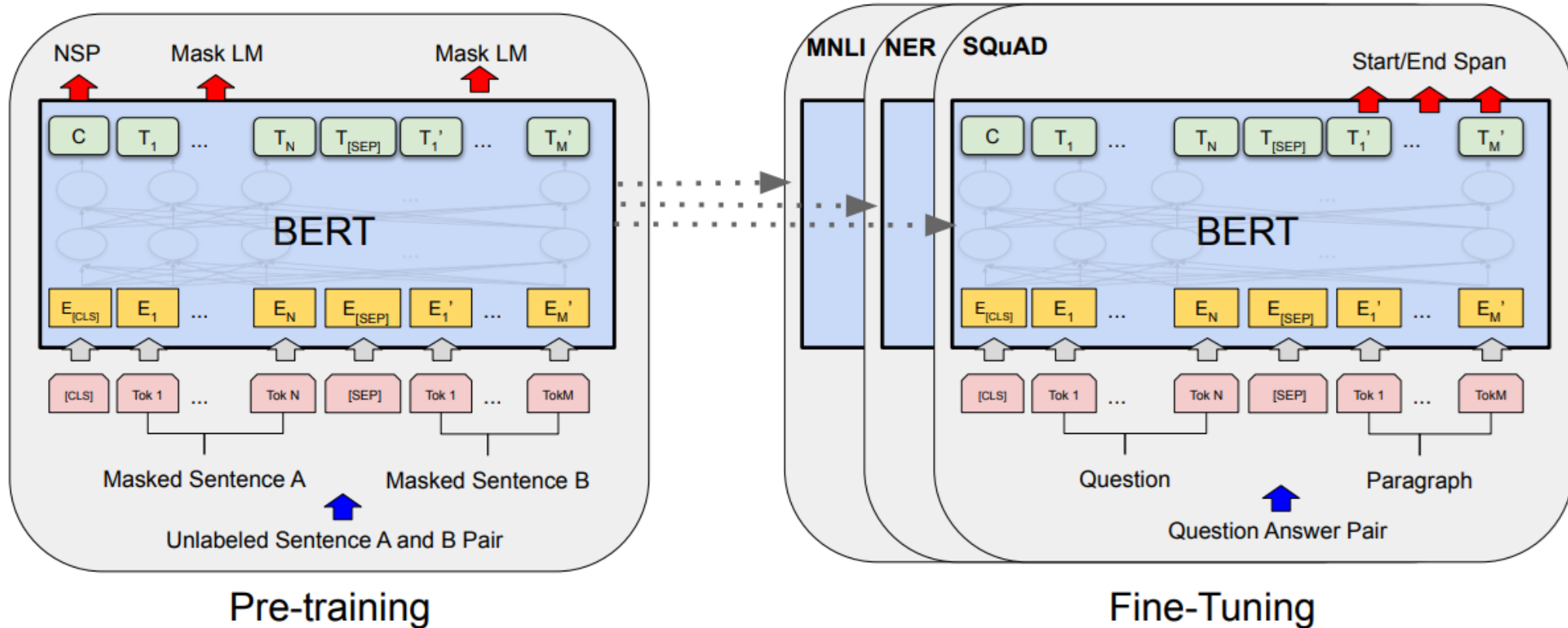


interact & give feedback

ref: H2020 easyRights project

# Functional Diagram

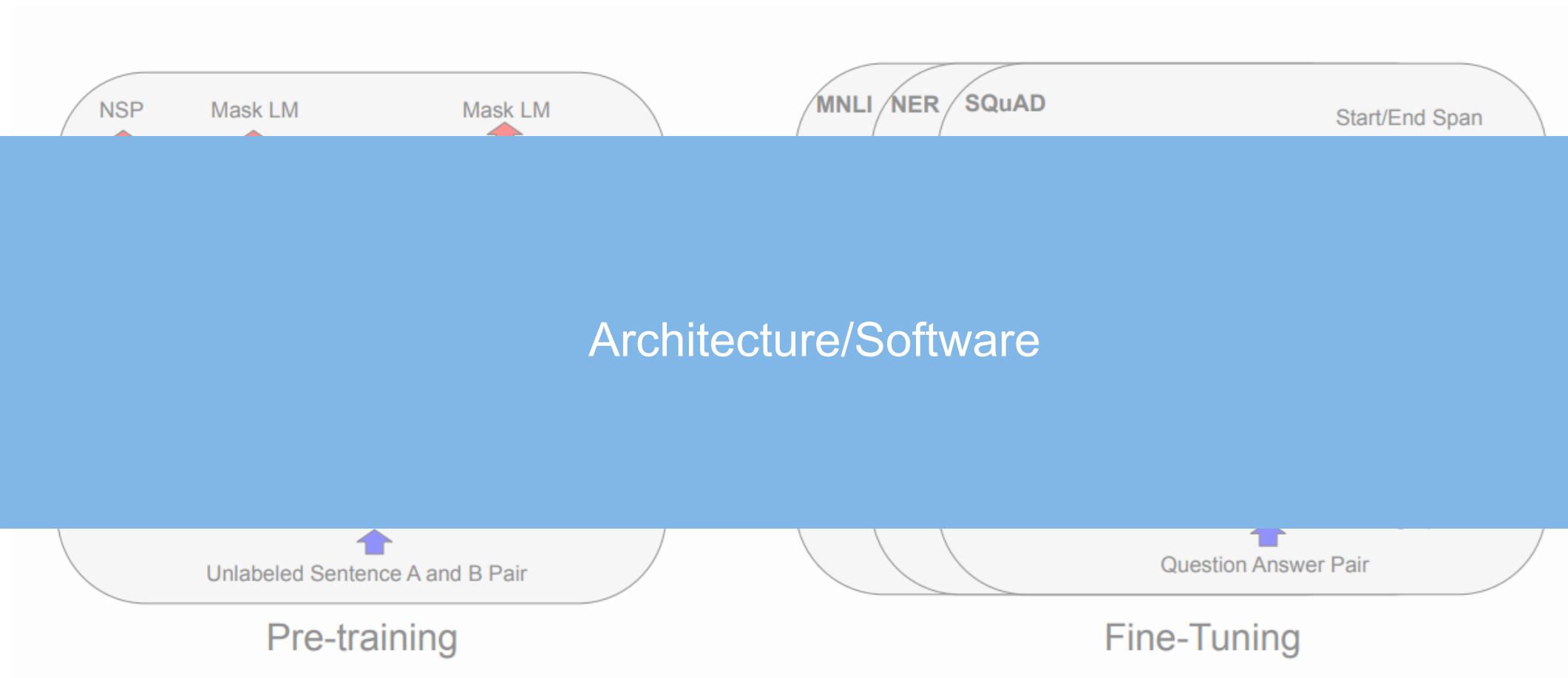
## Pillar 1



ref: BERT

# Functional diagram

## Pillar 1



# Manage

## Pillar 2

### Group activities in work packages

A **work package** is a building block of the work breakdown structure that allows the project management to define the steps necessary for completion of the work

A work package can be thought of as a sub-project, which, when combined with other work package units, form the completed project

Breaking down the work into work packages allows multiple teams to work simultaneously or sequentially on different components of the project

re. <https://ec.europa.eu/chafea/health/beneficiaries-corner/documents/factsheet-03.pdf>

# Manage

## Pillar 2

Work package is defined by a description, tasks, and outputs

### Description

Each work package aims to achieve one or more of the project objectives. The description of the work package should start with the statement of the project objective

The work package objectives define the methods, actions, and evaluation of the work package outputs and deliverables



# Manage

## Pillar 2

Work package is defined by a description, tasks, and outputs

### Task

It is the description of an activity where to list methods to reach the objective





# Manage

## Pillar 2

Work package is defined by a description, tasks, and outputs

### Output

It can be either a software, or a visual or a document





# Manage

## Pillar 2

Work packages, tasks, outputs have meaningful names to let the reader understand what is the work about



# Manage

## Pillar 2

|             |               |    |     |           |    |    |    |           |      |     |     |      |
|-------------|---------------|----|-----|-----------|----|----|----|-----------|------|-----|-----|------|
| WP N°       | 4             |    |     |           |    |    |    | WP Leader | P4   |     |     |      |
| WP title    | ADSP solution |    |     |           |    |    |    |           |      |     |     |      |
| Participant | 1             | 2  | 3   | 4         | 5  | 6  | 7  | 8         | 9    | 10  | 11  | 12   |
| Short name  | P1            | P2 | P3  | <u>P4</u> | P5 | P6 | P7 | P8        | P9   | P10 | P11 | P12  |
| PM          | 15            | 3  | 5.5 | 46        | 41 | -  | 6  | -         | 33.5 | 5.5 | 21  | 16.5 |
| Start month | 4             |    |     |           |    |    |    | End month |      |     | 36  |      |

### Objectives

This WP will focus on the development of ...

The activities will focus on ...

The specific objectives of this WP are to:

- Task1;
- Task2;
- Task3;
- Task4.

### Description of work

**T4.1 Task name (M4-M36) – Leader: P1 – Contributors: P4, P2, P1, P6.**

This task focuses...

**T4.2 ...**

T4.3 ...

T4.4 ...

### Deliverables:

**D4.1 Deliverable Name [P1] - Contributors: P4, P2, P1, P6. - (M12, M24, M36) - Description of the ...**

**D4.2 ...**

**Align** the task of the solution you are aiming to build to a known task of artificial intelligence (vision, language, ...)

For instance, you are building a solution to onboard users remotely only using a recorded video and a picture of the same user present in its electronic identification document

The task is about recognizing whether the user in the video is the same as in the picture

This is a task of **classification** (of computer vision)



# Stand on the shoulders of giants

## Pillar 3

The artificial intelligence community made it available a consistent number of **models** that can be taken, adapted via transfer learning, and used

This has been demonstrated being effective because:

- it avoids the cold start problem when developing a new solution
- it helps reasoning in terms of aligning a task to existing ones
- it inherits the capability of the model to perform over the task

# Workspace

## Pillar 3

Shared workspace for coding

Use a meaningful name for the workspace and start filling a README with the main objectives of the project

# Modular development

## Pillar 3

Structure the **development in modules** porting the names of the utilized diagram

Each module:

- has an owner of the development
- has needs (defined) inputs
- generates (expected) outputs

# Test & Scoring

## Pillar 3

Define a **testing environment** and make it reproducible multiple times

- define once the dataset with a portion of data to be utilized for training and one portion for development and another for testing
- define the metrics, for instance accuracy or F-measure
- utilize/develop a framework for scoring. Recall: keep it fixed throughout the execution of the project unless you want to repeat all experiments all times you do a change



# Communicate

## Pillar 4

Strategizing the communication is usually the output of a task(s)/work package

Take the stakeholder groups you defined in the design and communicate with them

Tailor the communication according to your stakeholder groups

Keep it alive throughout the entire project (from the beginning till the end)





# Communicate

## Pillar 4

**Visuals** work best for managers/clients/customers

**Spreadsheets** work for managers/project team

**Narratives of a technical document** work for project team and additional developers



# A good communication starts with a value proposition

## Pillar 4

A project has a mission (1 sentence)



Accelerate the world's transition to sustainable energy



# Thank you for your attention.

Questions?



# CONTACTS

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